Machine learning for the study of glacial-interglacial cycles

Using SINDy to reverse-engineering the proxy data, recovering (sparse) nonlinear equations that represents the dynamics of the system

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Proposed experiments

At least 2 experiments:

Experiment A - SINDy applied to PACCO BASE output.
To test the convergence of the method in the most favorable conditions.

Experiment D - SINDy applied to real proxy data.
¿Does cryosphere present hysteresis?

Optional experiments to assess if the results are robust

Experiment B - Experiment A + random noise.
To test the impact of the noise on the stability of the method and results.

Experiment C - Experiment A + random noise + non-equally spaced observations.
To test the impact of the observations dispersity.

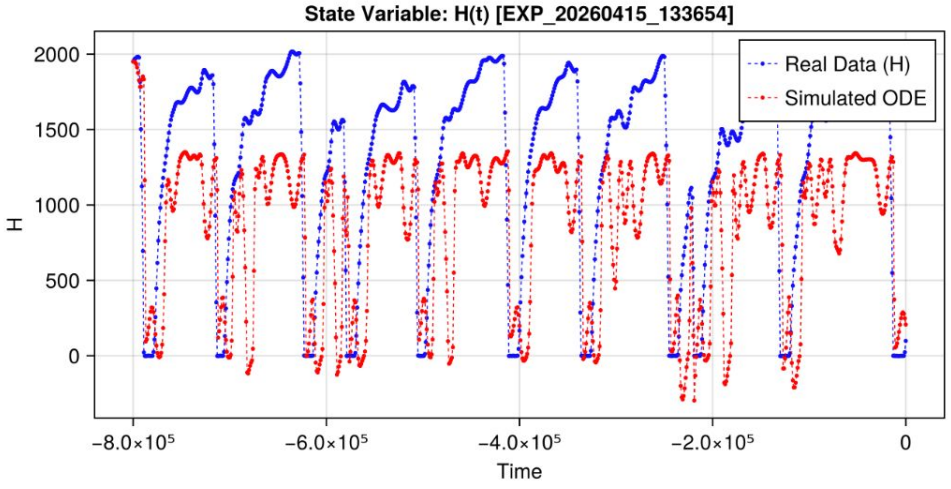
What Is PACCO?

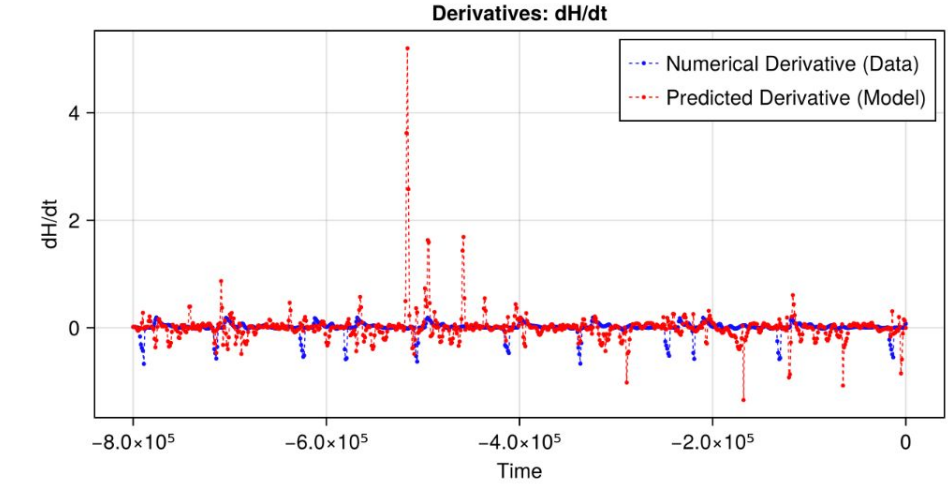
We already know!



Experiment A - Methodology

To see in the code





Experiment A - Partial results

Basis Degree: 4th Degree Polynomial

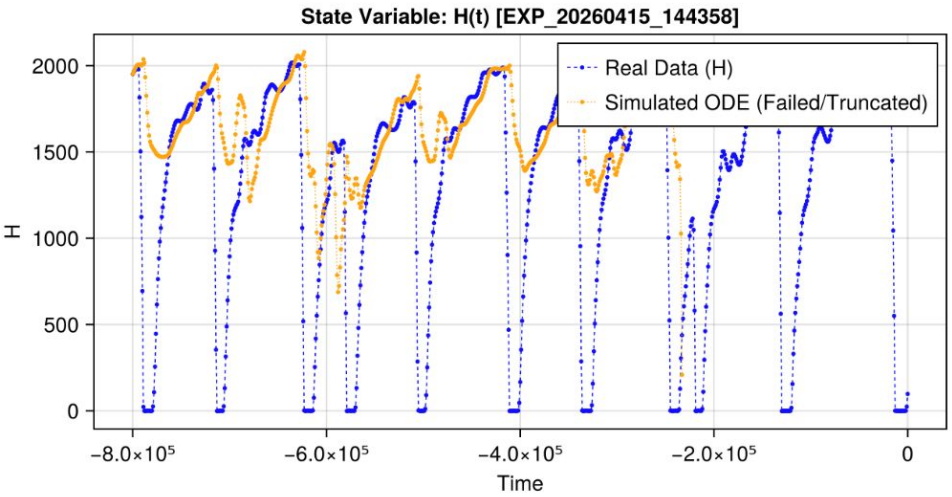
Solver Algorithm: ADMM (Alternating Direction Method of Multipliers)

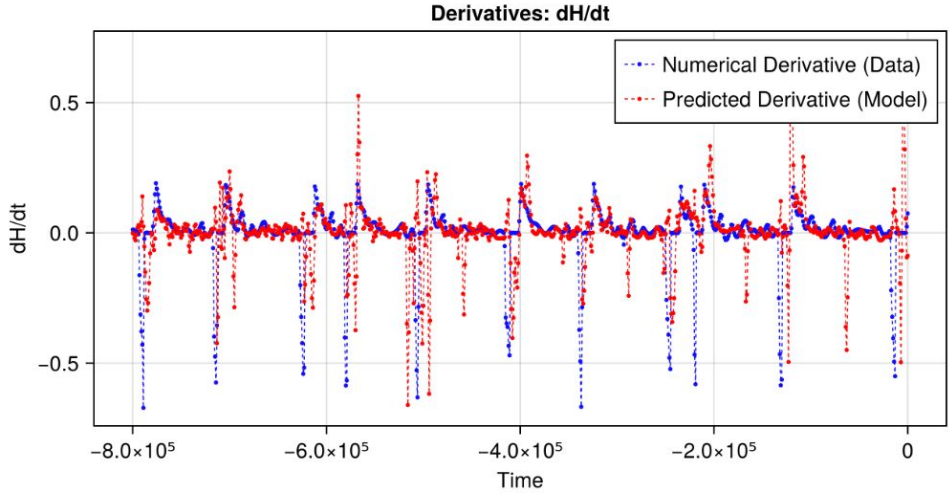
Model Accuracy (R2): -0.859

Discovered Equation:

dH/dt = p1 + p2 i + p6h + p14ch + p17c2 + p3i2 + p7ih + p9h2 + p11h3 + p15ch2 + p4i3 + p10i2h2 + p12ih3 + p13h4 + p16ch3 + p18c4 + p5i4 +p8i3h

Too complex.
dH/dt and H not well captured.





Experiment A - Partial results

Basis Degree: 5th Degree Polynomial

Solver Algorithm: ADMM (Alternating Direction Method of Multipliers)

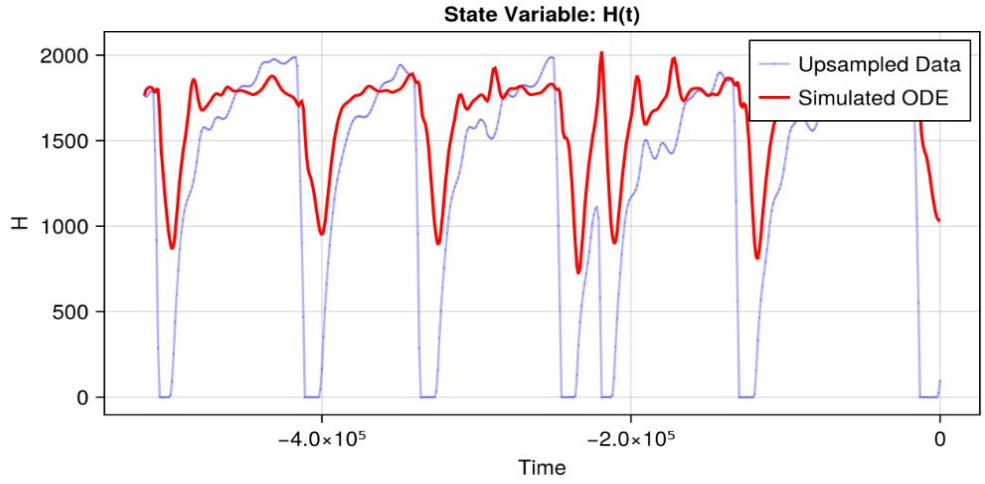
Model Accuracy (R^2): -7.437

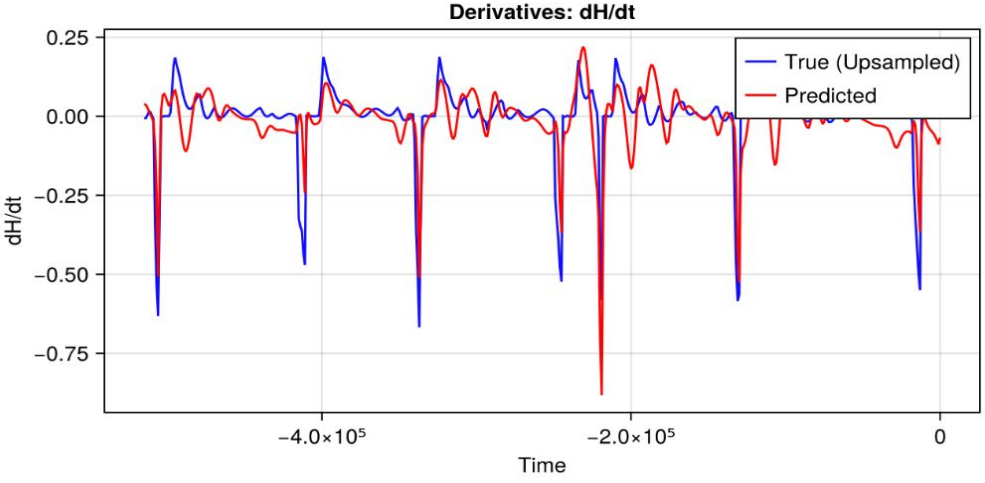
Degrees of Freedom (Parameters): 32

Discovered Equation:

dHd=p1i+p4h+p17ch+p2i2+p22c2+p5ih+p8h2+p12h3+p19ch2+p23c2i+p6i2h+p9ih2+p10i2h2+p14h4+p18ci2h+p24c2i2+p29c3h+p31c4+p11i3h2+p13i2h3+p15ih4+p16h5+p20cih3+p21ch4+p25c2i2h+p26c2ih2+p27c2h3+p28c3i2+p3i5+p30c3h2+p32c5+p7i4h

Much more complex and ODE explosion.
dH/dt and H better captured.





Experiment A - Best? result

(Timeframe: -515k to 0)

Basis Degree: 2nd Degree Polynomial

Solver Algorithm: STLSQ (Sequential Threshold Least Squares)

Model Accuracy (R^2): 0.462

Degrees of Freedom (Parameters): 10

Discovered Equation:
dH/dt = p1 + p2i + p4h + p7c + p10c2 + p3i2 + p5ih + p6h2 + p8ci + p9ch

Sparser, no ODE explotion.
dH/dt and H better captured.
REALLY REALLY UNSTABLE.

Experiment A - "Identified" problems

If basis is complex, the spikes are better resolved, but the ODE explodes.

If the basis is simple, the spikes are not well captured, but the ODE does not explodes.

Any idea?